

A common flag basis for the two-stage interchange

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The partial-swap batching lemma (`partial-swap-batching.tex`) compiles an edge of rank $a > m/2$ into $m - a$ corner transpositions and one aligned block, after one unimodular change of basis S_0 . A generic S_0 puts every corner in the open Bruhat cell, so the corner transpositions are reversed and each costs one singleton call. This note gives one rational basis in which the corners of every auxiliary and centre edge of the two-stage motif are lower triangular, so their transpositions form one contiguous run.

Runs. For an idempotent p of any rank, the lower–lower Bruhat involution of S_p is exactly $\{H_i \leftrightarrow D_j : (i, j) \text{ a rightmost pivot of } p\}$: the rank function of the blocks of rows $1..i$ and columns $j..2m$ of S_p is that of p on the $H \times D$ part and that of the identity elsewhere, for the H rows and the D rows alike. In particular S_p has exactly rank p transpositions, with no condition on rank p . A maximal run of pivots $(i, j), (i+1, j+1), \dots, (i+w-1, j+w-1)$ is one interchange of the contiguous chunks $H_{[i, i+w)}$ and $D_{[j, j+w)}$: the aligned case is PR #10's middle block, and shifted runs are already PR #10's corner block. An edge of rank $r \leq m/2$ is one child exactly when its $r \times r$ block (first r rows, last r columns) is lower triangular with nonzero diagonal; an edge of rank $r > m/2$ compiles to two children (the corner run and the aligned middle block of width $2r - m$) exactly when its $(m - r) \times (m - r)$ corner is lower triangular with nonzero diagonal. These are closed conditions, so they must hold through the structure of the basis.

Construction. Write $F = \mathbb{Q}^h$ with the dot product, and let $U, V \in GL_h(\mathbb{Q})$ be generic. For $0 \leq \beta < h$ choose R_β supported on $[0..\beta]^2$ with $R_{h-1}[h-1, h-1] = 0$, and C_β supported on $[\beta..h-1]^2$ with $C_\beta[\beta, \beta] = 0$ for $\beta < h-1$. The first h rows of S are the functionals $\lambda_\beta \leftrightarrow UR_\beta V^\top$ and the last h columns of S^{-1} are the vectors $\mu_\beta \leftrightarrow U^{-\top} C_\beta V^{-1}$ on $F \otimes F$. The pairing conditions $\text{tr}(R_i^\top C_j) = 0$ hold automatically for $i < j$ (disjoint

supports); for $i \geq j$ they are solved in triangular order through the slack entry $C_j[j, i]$, whose coefficient $R_i[j, i]$ is nonzero. Complete $S^{-1} = [Z \mid W \mid \mu]$ with $\lambda Z = I$ and W a basis of a complement of $\text{span } \mu$ inside $\ker \lambda$. Then S is rational; its denominators and minors add finitely many primes to the list the odd radix of the two-stage transfer avoids.

Lemma 1 (Lower-triangular corners). *For every line t , the $h \times h$ corners of $I \otimes P_t$ and of $P_t \otimes I$ in the basis S are lower triangular, with diagonal entries $(t^\top V r_i)(c_i^\top V^{-1} G t)$ and $(t^\top U \rho_i)(\gamma_i^\top U^{-1} G t)$, where r_i, c_i are the i -th rows and ρ_i, γ_i the i -th columns of R_i, C_i .*

Proof. For $\beta < \beta'$ the supports give $R_\beta^\top C_{\beta'} = 0$ and $R_\beta C_{\beta'}^\top = 0$ identically, so the (β, β') entries of both corners vanish. The diagonal entries are the stated products of linear forms, which are nonzero polynomials in (U, V) . $\square$

A lower-triangular corner has identity Bruhat permutation, so the corner pivots are $\sigma(l) = m - h + l$: one run.

Proposition 1 (Profiles). *On a common Zariski-open set of (U, V) and of the free entries of R_β, C_β :*

- *every selected auxiliary edge, in both stages, compiles to two children, of widths $m - 2h$ and h ;*
- *every rank- h centre edge is one child of width h ;*
- *for any idempotent π of rank $r > h/2$, the edges $\pi \otimes P_b$ and $P_b \otimes \pi$ compile to $h - r$ singletons and one block of width $2r - h$;*
- *every data-entrance corner (size $2h - 1$) is nonsingular.*

Proof. The first two items are Lemma 1. For the third, the block of $\pi \otimes P_b$ is $A_b \pi A_b^{-1} C_b$ with $A_b = \tilde{A}_b U^\top$, so its Bruhat permutation is that of the $h \times h$ idempotent $A_b \pi A_b^{-1}$, and the partial-swap lemma applies one level down, with a nonzero-polynomial corner condition in U (in V for the other stage). For the fourth, nonsingularity reduces to four intersection conditions (λ restricted to the null space has rank h ; the null space meets $\text{span } \mu$ trivially; the dual null space restricted to $\text{span } \mu$ has rank h ; the dual span meets $\text{span } \lambda$ trivially), each with an explicit witness ($U^\top t_1 = V^\top t_2 = e_0; e_0, e_{h-2}; e_{h-1}, e_{h-1}; e_{h-1}, e_1$), followed by the choice of W . The motif has finitely many edge classes, so the product of all these nonzero polynomials is nonzero on the irreducible parameter space, and a rational point avoids its zero set. $\square$

Saving. With the global-matching side circuit (`independent/two-stage-bit/gmside.py`, $R = 403248$ at $h = 47$) and the profiles above, the batched moment certifies $a_b = 13341/10^9$ at $h = 47$.

Checks. `flagbasis.py` builds S exactly (with generic coefficients; the statements are Zariski-open, so small coefficient ranges can hit the closed exceptional set) and computes the full Bruhat profile of every edge class modulo $2^{61} - 1$ (and exactly over $\mathbb{Q}$ for a sample at $h = 6$): all auxiliary lines in both stages, all centre edges and sampled data pairs at $h = 6, 7, 8$ match the proposition. At the working dimensions, `flag_existence.py` 47 (and 46) evaluates every nonvanishing condition at one integer point modulo $2^{61} - 1$: the corner diagonal factors and inner corners on every line, and both data-entrance factors on sampled pairs (the family is invariant under $GL_h \times GL_h$, which acts transitively on the pairs, so one nonzero value per condition suffices). A nonzero value modulo p at an integer point shows the polynomial is not identically zero. Negative controls: a dense generic basis gives singleton corners; PR #16's nested basis merges the stage-two class only.